

Detailed Curriculum — Phase 1: Project Technology Accelerator

10 Working Days • 2 Hours/Day • 20 Hours • Objective: prepare the team on the project-specific skills required before development begins.

Module 0 • Project Architecture & Solution Walkthrough

Topics

- Review of existing Foundry architecture
- Existing Ontology
- Current Workshop applications
- Existing APIs
- React application architecture
- OSDK architecture
- Project coding standards
- Sprint planning
- Q&A

Hands-on Lab: *Shared architecture walkthrough — no lab, discussion and Q&A (2 Hours)*

Module 1 • OSDK Architecture

Topics

- OSDK overview and SDK concepts
- Architecture and Object APIs
- Actions and Queries
- Authentication

Hands-on Lab: *Develop first OSDK application*

Module 2 • React Application Development

Topics

- React architecture, components and hooks
- Routing and responsive layout
- State management
- Material UI

Hands-on Lab: *Develop a mobile-responsive application*

Module 3 • React + OSDK

Topics

- Object queries, actions and forms
- Search and tables

- Authentication and deployment

Hands-on Lab: *Vehicle Management App*

Module 4 • Authentication

Topics

- Azure AD and Multipass
- Session and token flow

Hands-on Lab: *Azure Login*

Module 5 • Barcode Integration

Topics

- Camera scan and HID scanner
- Keyboard wedge and image capture

Hands-on Lab: *Scan → Lookup → Update Ontology*

Project Sprint Plan — Phase 2: Guided Project Development

20 Hours • Objective: guide the client team while they implement the application through structured sprint reviews, implementation walkthroughs, architecture discussions, and implementation mentoring — based on the shared architecture, not generic labs. Each sprint combines training, hands-on building, and review.

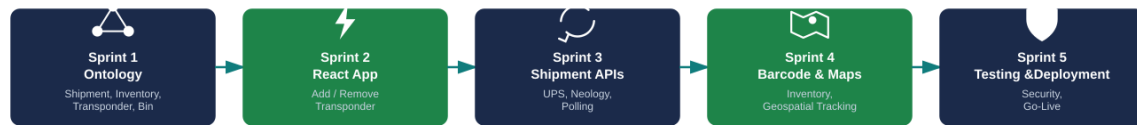


Figure 3 — Five-sprint guided development plan

Sprint	Week	Hours	Scope
Sprint 1	Week 3	4 Hours	Ontology — Shipment, Inventory, Transponder, Bin
Sprint 2	Week 4	4 Hours	React App — Add Transponder, Remove Transponder
Sprint 3	Week 5	4 Hours	Shipment APIs — UPS, Neology, Polling
Sprint 4	Week 6	4 Hours	Barcode, Inventory & Geospatial Visualization
Sprint 5	Week 7	4 Hours	Testing, Deployment, Security

Each sprint follows the same 4-hour structure: Architecture Discussion, Guided Development, and Review.

Sprint 4 Detail — Barcode, Inventory & Geospatial Visualization

In addition to camera and HID barcode scanning and inventory workflows, Sprint 4 explicitly covers the client's geospatial visualization requirement:

- Foundry Maps
- Geospatial Object Visualization
- Live Asset Tracking
- Vehicle Locations
- Shipment Tracking
- Map Layers
- Marker Styling
- Location Events

Phase 3: Project Mentoring

8 Hours • Weekly mentoring sessions running in parallel with guided development • Objective: sustain delivery momentum after the guided build, and transfer full ownership to the client's team.

- Weekly office hours
- Architecture review
- Code review
- Debugging support
- Performance tuning
- Deployment readiness
- Knowledge transfer

12. Deliverables

At the conclusion of the engagement, the client will receive:

All deliverables are intended as reusable reference assets to accelerate ongoing project implementation.

Client-Specific Deliverables

- Reviewed Transponder solution architecture
- React PWA reference implementation patterns
- Geospatial visualization reference implementation

Additional Deliverables

- React starter project
- OSDK sample application
- Authentication sample (Azure AD / Multipass)
- Barcode integration sample
- React component library
- API integration samples (UPS, Neology)
- Ontology extensions (Transponder, Shipment, Events, Bin)
- Architecture review notes
- Code review feedback
- Deployment checklist
- Best practices guide
- Project Implementation Checklist (environment, authentication, API, and deployment checklists)